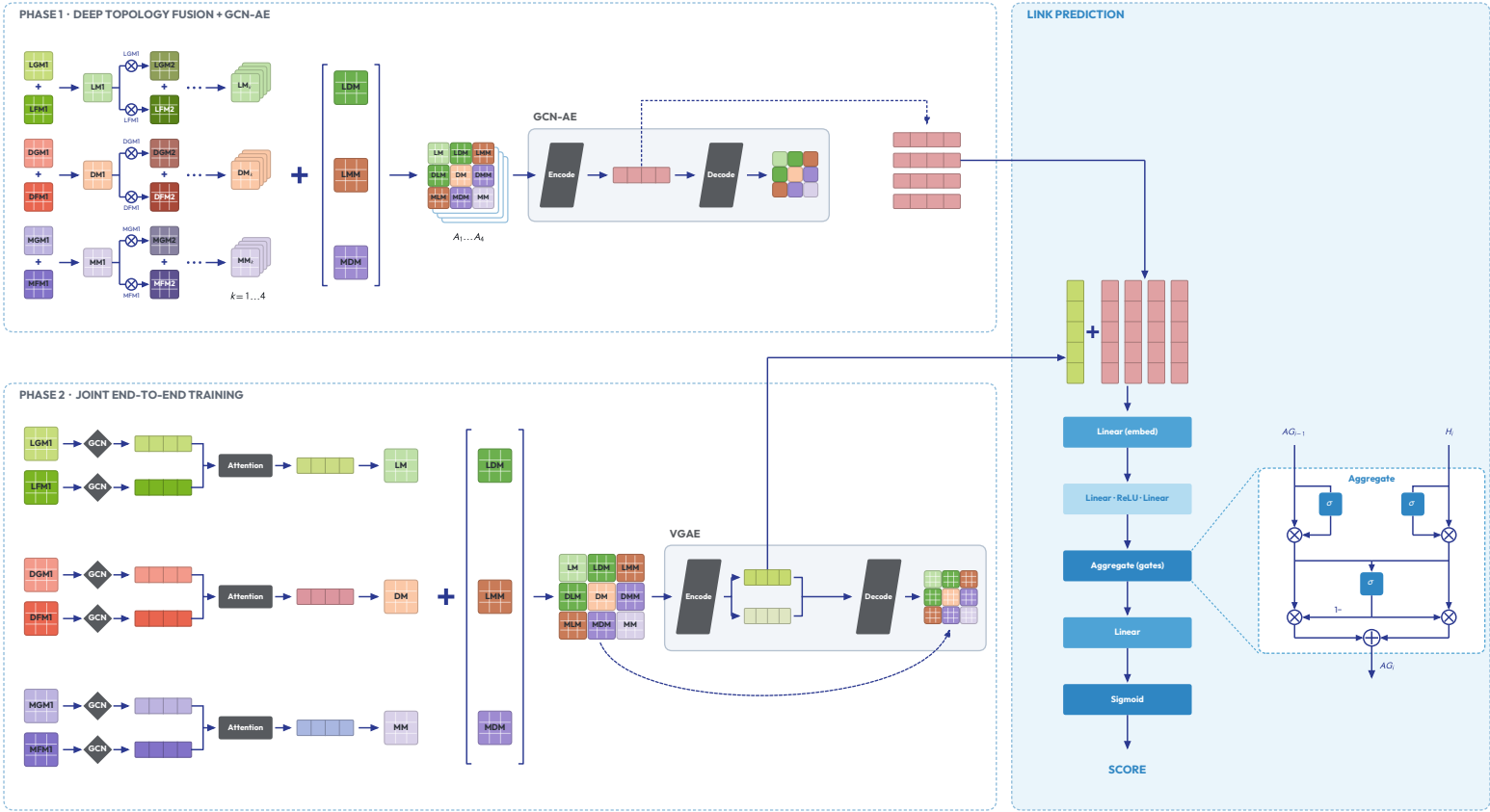
PHASE 1 · DEEP TOPOLOGY FUSION + GCN-AE
LGM1
LFM1
LM1
LGM2
LFM2
LM$_k$
DGM1
DFM1
DM1
DGM2
DFM2
DM$_k$
MGM1
MFM1
MM1
MGM2
MFM2
MM$_k$
$k = 1 \ldots 4$
LDM
LMM
MDM
LM
LDM
LMM
DLM
DM
DMM
MLM
MDM
MM
$A_1 \ldots A_4$
GCN-AE
Encode
Decode
LINK PREDICTION
PHASE 2 · JOINT END-TO-END TRAINING
GCN
Attention
VGAE
Linear (embed)
Linear · ReLU · Linear
Aggregate (gates)
Linear
Sigmoid
SCORE
AG_{l-1}
H_l
Aggregate
σ
1−
AG_l